Cloud Transformation Strategy

A High-Performance Architecture Blueprint for CVS

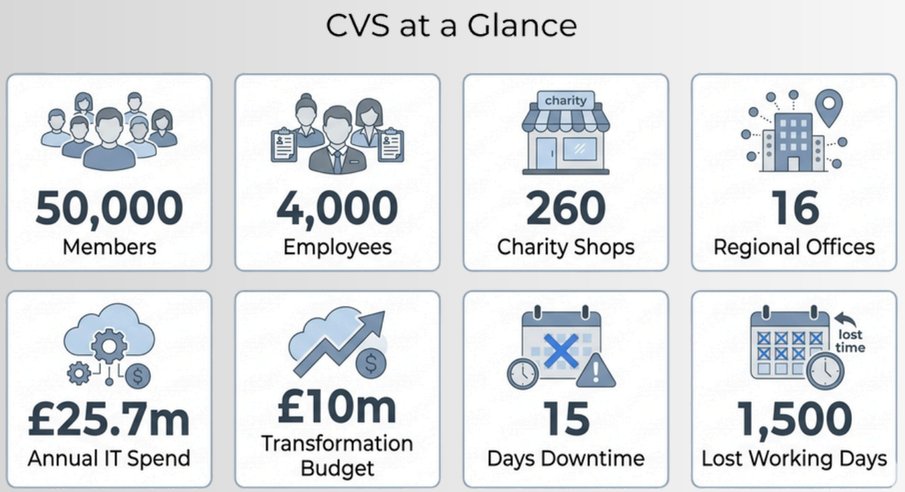
UNIVERSITY OF THE WEST OF ENGLAND, BRISTOL



Drivers for Strategic Change

Christian Volunteer Service (CVS) requires rapid modernization to secure its £96M+ donation pipeline across 260 nationwide shops.

- **Unsustainable IT Overhead:** £25.7m annual spend weighed down by deep fragmentation (e.g., 5 disparate payroll systems).
- **Critical Operational Risk:** 15 days of legacy system downtime directly translates to 1,500 lost working days.
- **Strategic Shift Required:** Board mandate to transition from high maintenance CapEx to predictable, scalable OpEx (Armbrust et al., 2010).
- **Untapped Revenue:** Current systems lack the capability to execute a robust national e-commerce retail strategy.



SWOT: Current IT Estate

STRENGTHS & OPPORTUNITIES (POSITIVE)

Internal Strengths

Deep institutional operational knowledge, high local responsiveness across 16 regional offices, and rich existing legacy donor datasets.

External Opportunities

Leveraging modern cloud platforms to unify data, implement predictive donor targeting, and launch high value e-commerce capabilities.

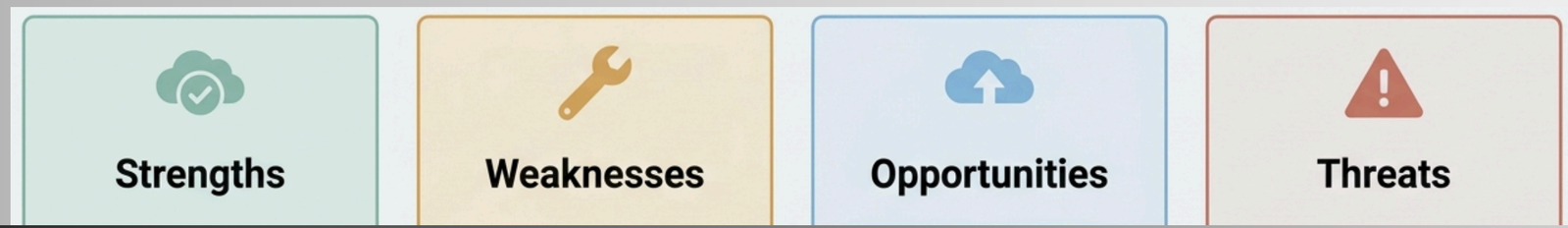
WEAKNESSES & THREATS (NEGATIVE)

Internal Weaknesses

Massive technical debt (£8.5m annual software fragmentation), crippling CRM unreliability, and IT staff paralyzed by maintenance duties.

External Threats

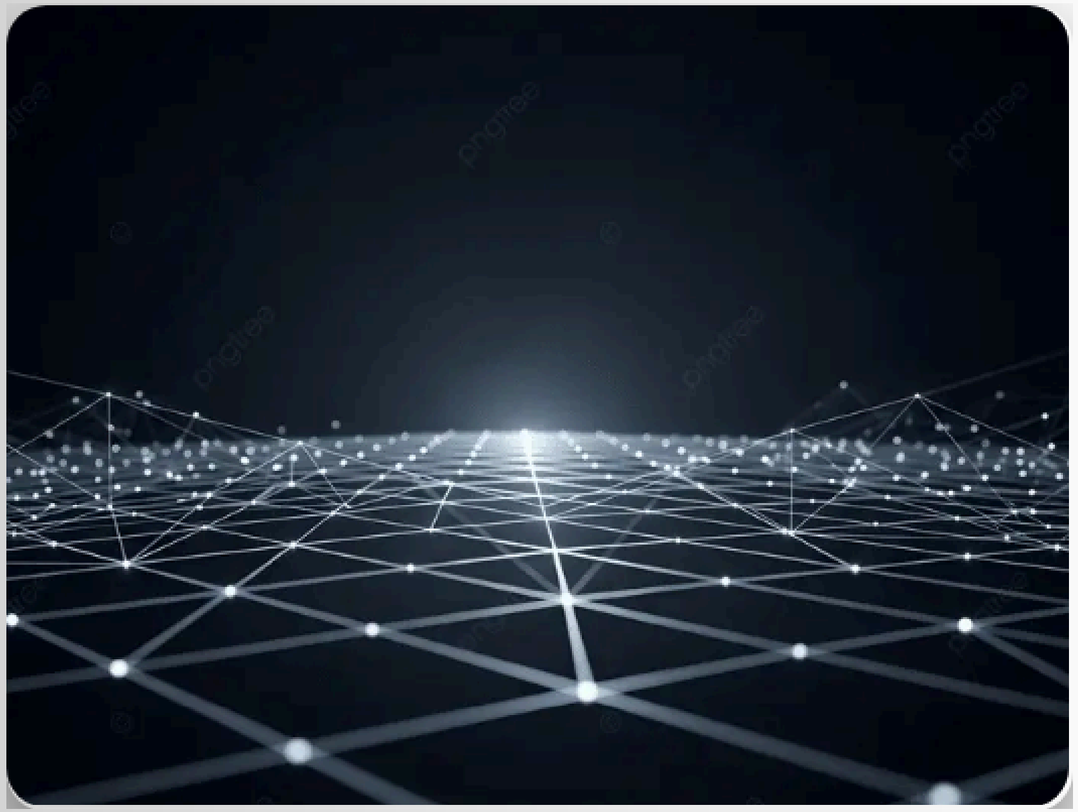
Imminent service failure disrupting critical fundraising continuity, elevated cybersecurity risks in unpatched systems, and staff union resistance.



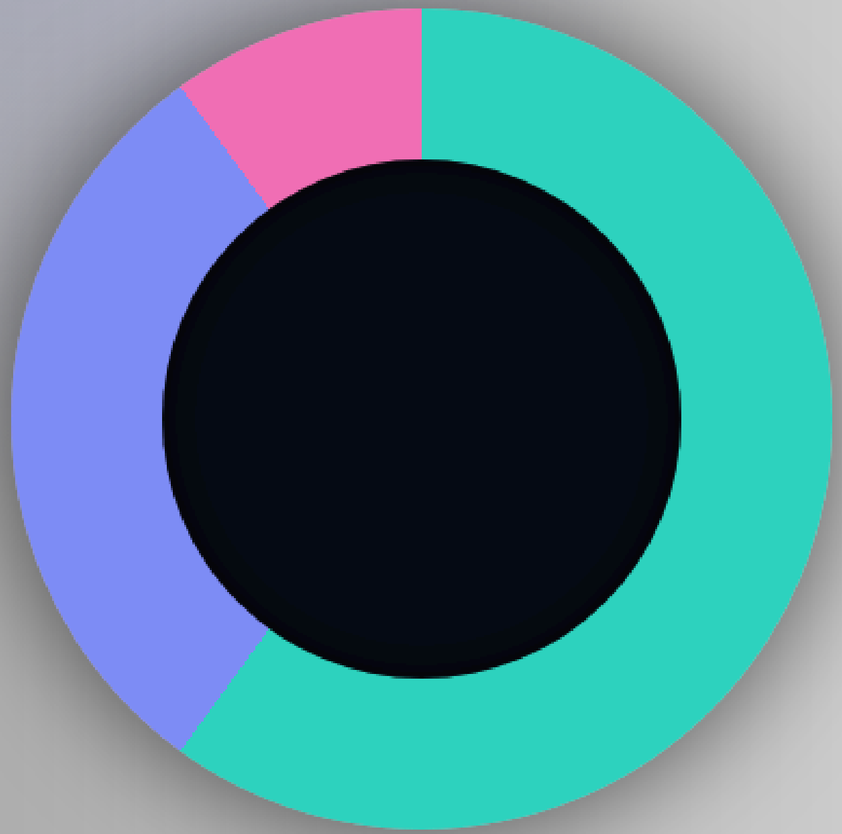
Target Hybrid Architecture

Our proposed architecture prioritizes business enablement, agility, and absolute security through a strictly governed hybrid model.

- **Standardized SaaS:** Unifying CRM, HR, Payroll, and collaborative productivity to eliminate regional fragmentation entirely.
- **Strategic PaaS:** Deploying centralized data lakes, advanced integration APIs, and providing the foundation for e-commerce.
- **Restricted IaaS:** Kept to an absolute minimum; functioning solely as a transitional container for immovable legacy systems.



Recommended Cloud Service Mix



→ SaaS (60%) - Core Applications

→ PaaS (30%) - Data & Innovation

→ IaaS (10%) - Legacy Transition

- By heavily weighting **Software as a Service (SaaS)**, CVS accelerates deployment and dramatically lowers overhead (Mell & Grance, 2011).
- **Platform as a Service (PaaS)** drives centralized analytics, while **Infrastructure as a Service (IaaS)** acts strictly as a temporary bridge, preventing the re-import of technical debt.

Why Microsoft Azure?

Enterprise Alignment

Azure provides seamless, native integration across Microsoft 365, Entra ID, and Dynamics 365, ensuring robust identity governance for distributed charity staff.

Financial Predictability

Transforms inefficient, unpredictable annual CapEx into scalable, highly transparent OpEx, maximizing the value of the £10M transformation budget.



Mission Velocity

By liberating IT personnel from mundane database maintenance, highly skilled resources are reallocated to donor intelligence and strategic execution.

3-Year Transformation Budget

Investment Pillar	Year 1	Year 2	Year 3	Total
SaaS Implementation (CRM, HR, Payroll)	£1.00m	£0.80m	£0.30m	£2.10m
Data Migration & Core Systems Integration	£0.90m	£0.70m	£0.30m	£1.90m
PaaS Analytics & E-commerce Platform	£0.50m	£0.70m	£0.50m	£1.70m
Architecture, Training & Change Mgmt	£1.05m	£0.65m	£0.40m	£2.10m
Cybersecurity, Identity & Backup Tools	£0.40m	£0.30m	£0.25m	£0.95m
Transitional IaaS & Contingency (6%)	£0.50m	£0.45m	£0.30m	£1.25m
Total Allocated Investment	**£4.35m**	**£3.60m**	**£2.05m**	**£10.00m**

Unlocking Strategic Value

1,500

LOST DAYS ELIMINATED

Through legacy platform retirement.



Information as a Service (INaaS)

INaaS transforms raw data from a siloed maintenance burden into an accessible, high velocity organizational asset.

- **Unified Data Layer:** Eradicating disjointed Excel sheets across 16 regions by centralizing donor and retail data via Azure Data Factory.
- **Role Based Analytics:** Equipping regional managers with real time Power BI dashboards.
- **Predictive Intelligence:** Harnessing historical data to precisely segment donor campaigns and forecast high yield fundraising trends.

BPaaS Implementation Roadmap

Business Process as a Service (BPaaS) transcends simple cloud hosting it enforces the standardization of complex, end-to-end organizational workflows globally.

Year 1: HR & Payroll

Consolidate the 5 disparate regional payroll packages. Standardize the employee lifecycle, onboarding, and compliance globally.

Year 2: Donor Workflows

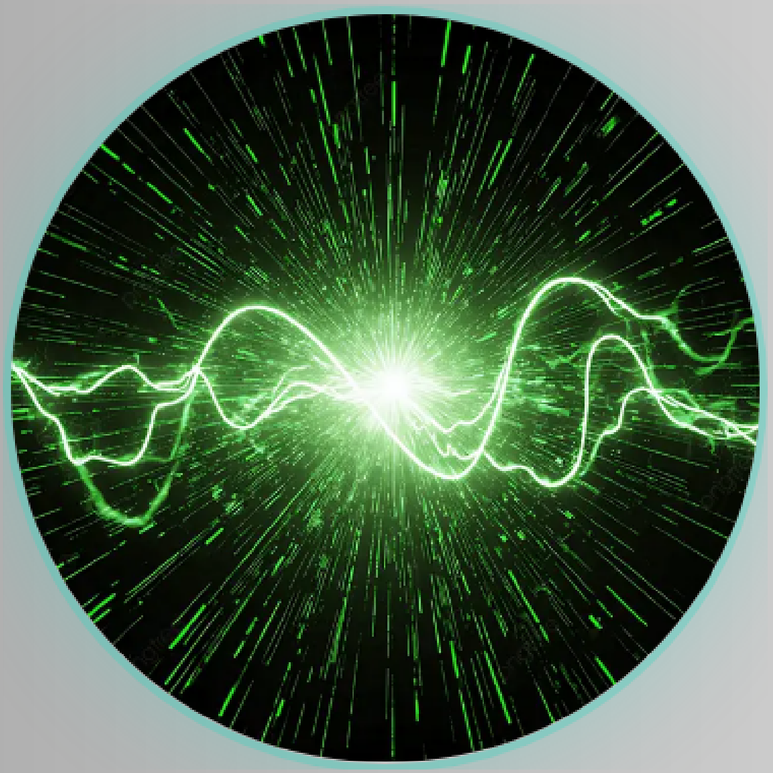
Automate intricate stewardship processes. Integrate data driven campaign segmentation and execution across all digital channels.

Year 3: E-Commerce

Launch the national retail validation process. Streamline photo uploads, central expert pricing, and complex outbound logistics.

Year	BPaaS Priority	Expected Benefit
Year 1	Payroll and HR workflow standardization	Lower fragmentation, better compliance
Year 2	Donor communication and fundraising automation	Enhanced accountability and responsibility
Year 3	National e-commerce business process rollout	New capability of generation of income

Sustainability Impact



Driving Environmental Integrity

Transitioning from a heavily fragmented, localized hardware estate to hyperscale cloud infrastructure fundamentally optimizes energy consumption (IEA, 2024).

- **Carbon Negative Mandate:** Microsoft Azure aligns seamlessly with strict global ESG commitments, aiming for carbon negative operations by 2030 (Microsoft, 2024).
- **Scope 3 Eradication:** Halting the continuous cycle of localized hardware purchasing and disposal drastically slashes supply chain emissions.
- **GreenOps Governance:** Leveraging native tooling to monitor emissions, rigorously right size workloads, and destroy redundant "dark data".

Risk Mitigation Strategies

Cybersecurity & Misconfiguration Vulnerabilities Cloud migrations expose legacy flaws. We strictly enforce Zero Trust baselines, mandatory Entra ID MFA, and continuous threat monitoring (ENISA, 2021).

Data Migration Integrity Safeguarding the 6.4GB of legacy donor data is non negotiable. Mitigated through aggressive data cleansing, parallel pilot runs, and stringent rollback protocols.

Staff Resistance & Change Fatigue Union concerns regarding redundancy are significant. Addressed via absolute transparency, comprehensive cloud upskilling for IT staff, and early stakeholder buy in.

Scope Creep & Budgetary Overrun The £10M budget limit is strictly enforced through FinOps practices, a phased delivery roadmap, and aggressive avoidance of proprietary vendor lock in.

Key References

- Armbrust, M., Fox, A., Griffith, R., Joseph, A.D., Katz, R., Konwinski, A., Lee, G., Patterson, D., Rabkin, A., Stoica, I. and Zaharia, M. (2010) 'A view of cloud computing', *Communications of the ACM*, 53(4), pp. 50–58.
- ENISA (2021) *Cloud Security for SMEs and Public Administration*. European Union Agency for Cybersecurity.
- IEA (2024) *Data Centres and Data Transmission Networks*. International Energy Agency.
- Kruchten, P., Nord, R.L. and Ozkaya, I. (2012) 'Technical debt: From metaphor to theory and practice', *IEEE Software*, 29(6), pp. 18–21.
- Marston, S., Li, Z., Bandyopadhyay, S., Zhang, J. and Ghalsasi, A. (2011) 'Cloud computing - The business perspective', *Decision Support Systems*, 51(1), pp. 176–189.
- Mell, P. and Grance, T. (2011) *The NIST Definition of Cloud Computing*. NIST Special Publication 800-145. Gaithersburg: National Institute of Standards and Technology.
- Microsoft (2024) *Microsoft Sustainability*. Corporate Environmental Report.
- NIST (2020) *Security Strategies for Microservices-based Application Systems*. National Institute of Standards and Technology.

Thank You !...